



Queries On World DB

The world database contains information about countries, cities, and languages, and it's a great resource for practicing SQL queries. Below are some example assignments with queries to help you get started:

1. Basic Select Queries

List all countries and their continents.

```
SELECT Name, Continent FROM country;
```

List all cities in Japan.

```
SELECT Name FROM city WHERE CountryCode = 'JPN';
```

2. Filtering Data

Find all countries with a population greater than 100 million.

```
SELECT Name, Population FROM country WHERE Population > 100000000;
```

Find all cities with a population between 1 million and 5 million.

```
SELECT Name, Population FROM city WHERE Population BETWEEN 1000000 AND 5000000;
```

3. Sorting Data

List the top 10 most populous countries.

```
SELECT Name, Population FROM country ORDER BY Population DESC LIMIT 10;
```

List all cities in China ordered by population in descending order.

```
SELECT Name, Population FROM city WHERE CountryCode = 'CHN' ORDER BY Population DESC;
```

4. Aggregate Functions

Calculate the total population of all countries in Asia.

```
SELECT SUM(Population) AS TotalPopulation FROM country WHERE Continent = 'Asia';
```

Find the average population of cities in the United States.

```
SELECT AVG(Population) AS AveragePopulation FROM city WHERE CountryCode = 'USA';
```

5. Group By

List the number of countries in each continent.

```
SELECT Continent, COUNT(*) AS NumberOfCountries FROM country  
GROUP BY Continent;
```

Find the average life expectancy per continent.

```
SELECT Continent, AVG(LifeExpectancy) AS AverageLifeExpectancy FROM country GROUP  
BY Continent;
```

6. Joins

List all cities along with their country name.

```
SELECT city.Name AS CityName, country.Name AS CountryName  
FROM city  
JOIN country ON city.CountryCode = country.Code;
```

Find all languages spoken in each country.

```
SELECT country.Name AS CountryName, countrylanguage.Language  
FROM countrylanguage  
JOIN country ON countrylanguage.CountryCode = country.Code;
```

7. Subqueries

Find the country with the highest population.

```
SELECT Name, Population FROM country WHERE Population = (SELECT MAX(Population)  
FROM country);
```

List cities that have a population greater than the average population of all cities.

```
SELECT Name, Population FROM city WHERE Population > (SELECT AVG(Population) FROM city);
```

8. Using String Functions

Find all countries whose names start with 'A'.

```
SELECT Name FROM country WHERE Name LIKE 'A%';  
Find all cities whose names contain 'York'.
```

```
SELECT Name FROM city WHERE Name LIKE '%York%';
```

9. Updating Data

Increase the population of all cities in Japan by 10%.

```
UPDATE city  
SET Population = Population * 1.10  
WHERE CountryCode = 'JPN';
```

Change the continent of Australia to 'Oceania'.

```
UPDATE country SET Continent = 'Oceania' WHERE Name = 'Australia';  
10. Deleting Data
```

Delete all cities with a population less than 500.

```
DELETE FROM city WHERE Population < 500;  
Remove all languages spoken in only one country.
```

```
DELETE FROM countrylanguage WHERE CountryCode IN (  
  SELECT CountryCode FROM (  
    SELECT CountryCode FROM countrylanguage GROUP BY CountryCode HAVING  
    COUNT(Language) = 1  
  ) AS temp  
);
```

Queries used in Dashboard application

```
-- 1. Query to get the total world population  
SELECT SUM(Population) AS TotalPopulation FROM country;
```

```
-- 2. Query to get the top 10 most populated countries  
SELECT Name, Population FROM country ORDER BY Population DESC LIMIT 10;
```

```
-- 3. Query to get the top 10 least populated countries  
SELECT Name, Population FROM country ORDER BY Population ASC LIMIT 10;
```

```
-- 4. Query to get the total number of countries in the database  
SELECT COUNT(*) AS TotalCountries FROM country;
```

```
-- 5. Query to get the average population of all countries  
SELECT AVG(Population) AS AveragePopulation FROM country;
```

```
-- 6. Query to get the total population by continent  
SELECT Continent, SUM(Population) AS TotalPopulation FROM country GROUP BY Continent;
```

```
-- 7. Query to get countries with a population greater than a specified value (replace X with a  
number)  
SELECT Name, Population FROM country WHERE Population > X ORDER BY Population  
DESC;
```

```
-- 8. Query to get countries with a population less than a specified value (replace X with a  
number)  
SELECT Name, Population FROM country WHERE Population < X ORDER BY Population  
ASC;
```

-- 9. Query to get the population of a specific country (replace 'CountryName' with actual name)
SELECT Name, Population FROM country WHERE Name = 'CountryName';

-- 10. Query to get the official languages spoken in a country (replace 'CountryName' with actual name)
SELECT cl.Language FROM countrylanguage cl
JOIN country c ON cl.CountryCode = c.Code
WHERE c.Name = 'CountryName';

-- 11. Query to get the total number of languages spoken worldwide
SELECT COUNT(DISTINCT Language) AS TotalLanguages FROM countrylanguage;

-- 12. Query to get the top 10 most spoken languages worldwide
SELECT Language, COUNT(*) AS CountryCount FROM countrylanguage GROUP BY
Language ORDER BY CountryCount DESC LIMIT 10;

-- 13. Query to get countries and their respective GDP per capita (if GDP is stored in GNP field)
SELECT Name, GNP/Population AS GDPPerCapita FROM country ORDER BY GDPPerCapita
DESC;

-- 14. Query to get the average life expectancy of all countries
SELECT AVG(LifeExpectancy) AS AverageLifeExpectancy FROM country;

-- 15. Query to get the countries with the highest life expectancy
SELECT Name, LifeExpectancy FROM country ORDER BY LifeExpectancy DESC LIMIT 10;

-- 16. Query to get the countries with the lowest life expectancy
SELECT Name, LifeExpectancy FROM country ORDER BY LifeExpectancy ASC LIMIT 10;

-- 17. Query to get information about a specific city (replace 'CityName' with actual name)
SELECT * FROM city WHERE Name = 'CityName';

-- 18. Query to get the largest cities by population
SELECT Name, Population FROM city ORDER BY Population DESC LIMIT 10;

-- 19. Query to get the total number of cities in the database
SELECT COUNT(*) AS TotalCities FROM city;